Bridging Arctic Command Structure Seams to Support Allied Multi-Theater Defense

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Executive Summary

In an evolving threat environment defined by great-power competition, the Arctic has reemerged as a strategic theater central to the defense of North America and Europe. Growing civil-military cooperation between Russia and China, including in the Pacific Arctic, and strategic connections between the Arctic, North Atlantic, Baltic, and North Pacific reinforce the need to avoid compartmentalizing Arctic challenges into subregional problem sets—a tendency that adversaries can exploit. The potential for parallel or connected multi-vector crises with direct homeland defense implications increase the need for closer cooperation between the United States and its allies.

Allied posture in the High North directly affects North American warning, reinforcement, and escalation management, just as shifts in US global priorities shape the credibility of deterrence in Europe's north. This interdependence elevates the importance of interfaces among US combatant commands, NORAD, NATO, and key allied national structures. As these arrangements evolve, the central challenge lies in aligning authorities, information, and response options quickly enough to enable coherent action across domains and theaters.

This is particularly important as adversaries use hybrid, conventional, and nuclear toolkits to undermine the Western security architecture and unity of purpose by targeting vulnerabilities, and exploiting gaps and seams between institutions, agencies, commands, and systems that can delay decision-making, fragment situational awareness, or complicate coordination.

Regional seams are most acute where operational challenges span more than one AOR or jurisdiction but planning and decision authorities remain partitioned. In Arctic-relevant scenarios, these seams typically appear in three areas: across national AOR boundaries, where warning and response pathways must be synchronized; at interfaces between national and allied or binational structures, where information-sharing and command relationships must be pre-aligned; and within alliance command arrangements, where collective-defense requirements may not fully align with operational boundaries.

Functional seams further compound these challenges. They emerge across interagency, civil-military, and public–private boundaries, as well as at the intersection of data and decision-making systems, where effectiveness depends on the timely fusion, processing, and dissemination of information. They also arise when cross-domain effects are required, but authorities, processes, and data remain organized in stovepipes.

Seams are an enduring feature of complex command arrangements. The critical issue is whether they are anticipated and managed in ways that reflect coherent plans, authorities, and decision timelines. While consolidation may shorten formal chains of command, it does not eliminate seams—it redistributes them. If accompanied by reduced regional presence, smaller staffs, or weaker liaison networks, such changes can create new vulnerabilities that adversaries can exploit. In the Arctic, where distance, environmental conditions, and sparse infrastructure amplify delays, limit situational awareness and operational posture, these risks are particularly acute. They are further exacerbated when reforms outpace the planning, exercising, and

transition management needed to make them operationally credible, potentially weakening cohesion across allied structures.

Rather than eliminating seams, the objective is to ensure they do not become strategic liabilities. A command architecture capable of absorbing friction and maintaining political and military coherence, provides the best path in denying adversaries opportunities to exploit physical, procedural, or cognitive gaps.

This report identifies several priority lines of effort to minimize seams across US and allied structures:

- Improving clarity and coordination across the alliance, including clear communication of US commitments, expectations, and more predictable coordination mechanisms to enable realistic planning and unity of effort.
- Strengthening intelligence-sharing through more consistent releasability standards, pre-approved disclosure pathways, and trusted channels that reduce delays in shared situational awareness.
- Enhancing cross-command integration through reinforced liaison networks and institutionalized coordination across key US-NATO command interfaces.
- Expanding exercises to operationalize coordination, testing cross-AOR integration, command transitions, and the prioritization of global enablers under realistic conditions.
- Investing in resilient, multi-domain situational awareness and decision-making capabilities, including shared sensing and communications infrastructure and the careful integration of AI-enabled tools, while preserving the central role of human judgment.

Taken together, Arctic defense will remain defined by interdependence across regions, domains, and institutions. The decisive factor is not the elimination of seams, but the ability to anticipate, understand, and manage them effectively under conditions of simultaneity and uncertainty. In this context, resilience, coordination, and operational coherence are essential to maintaining credible deterrence and defense in the Arctic.

Introduction

The Arctic[1] has reemerged as a strategic theater central to the defense of North America and Europe, where the security interests of the United States, Canada, and European partners directly align, underpinned by complementary, interdependent, and mutually reinforcing security postures. NATO and its Nordic allies, with essential support from the United States, are critical to North Atlantic defense against a resurgent Russia. Allied posture and capabilities in the High North[2] also support North American continental defense, which is a key priority for the United States in a two-nuclear near-peer threat environment.[3]

The administration of President Donald J. Trump has prioritized the defense of the Western Hemisphere and the First Island Chain in the Indo-Pacific[4], adding pressure to reallocate resources and personnel from Europe despite Russia's persistent threat to Euro-Atlantic security. This shift has direct implications for defense and deterrence in the Arctic and its strategically interconnected regions, including the Baltic Sea and North Pacific.

Deepening coordination among Washington's adversaries and the potential for parallel or connected multi-vector crises that can rapidly generate homeland defense implications demand greater cooperation among the United States and its allies.

This takes on greater urgency in a context in which adversaries increasingly rely on cyber, space, influence operations, and gray-zone activities, creating overlapping and at times competing requirements for the United States and its allies,[5] while seeking to undermine the Western security architecture and unity of purpose by targeting vulnerabilities, exploiting gaps (shortfalls in capability, capacity, or coverage) and seams (organizational, conceptual, and procedural boundary lines) between institutions, agencies, commands, and systems that can delay decision-making, fragment situational awareness, or complicate coordination.[6]

Collaboration is especially relevant in the Arctic, where US and allied resources and postures are limited and where a challenging operational environment exacerbates gaps and complicates response options. In light of the deep interconnections between the Arctic, the North Atlantic, and Pacific, the United States should seek to develop an integrated posture that supports situational awareness and response capacity matched to threats across the conflict spectrum, even under conditions of simultaneity, while minimizing the potential for escalation. To achieve this, the United States must draw on the unique capabilities and insights of regional allies, optimize information-sharing, and support integrated command and control (C2). As the United States and US Northern Command (USNORTHCOM) assess the force posture needed to support national security objectives in the Arctic, these considerations are critical. Where requirements go beyond national jurisdictions, they demand close collaboration and coordination with allies.[7]

At a time when the United States is seeking to adapt its national military posture and reduce the number of US personnel within NATO[8], this report assesses where persistent or emerging seams and resulting gaps affecting Arctic defense planning and implementation—both on the

national and multinational level—impede adequate responses in a security environment defined by simultaneity and technological advancement. It examines the critical role of allies and partners in North American continental defense as Washington presses allies to take on more responsibility in the European theater.[9]

Previous studies have identified the need for greater coordination across US combatant commands (COCOMs) with responsibilities for defense in the Arctic and touched on overlapping and shared structures between the US and its allies.[10] This report seeks to bolster that research by discussing the current and prospective boundary points between the regionally relevant command structures, with a special focus on the interface between the United States, NORAD, and NATO. This is especially relevant because regional requirements, as well as US and NATO command structures and mission sets, are evolving. Relying on open source information, including strategy documents and analysis, anonymized interviews, and track 1.5 discussions with current and former US and allied officials and experts, the central question this report seeks to answer is: What steps can the United States and its NATO allies take to reduce exploitable seams that could hinder coherent Arctic and North American continental defense, especially under conditions of cross-theater simultaneity?

The Evolving Threat Landscape

The Arctic, which played a central role in US and transatlantic deterrence and defense during the Cold War, has reemerged as a critical theater in the era of great-power competition linking Northern Europe and the North Atlantic to the defense of North America.[11] Although the North American and European Arctic face distinct operational conditions and requirements, the threat landscape across these regions is deeply interconnected.

Over-the-pole trajectories continue to present the primary vector for missile threats aimed at North America from Russia, China, and North Korea.[12] At the same time, the European High North and North Atlantic approaches remain critical to European defense, US security, and escalation management against Russia.[13]

A changing global threat landscape, coupled with environmental[14] and technological developments, translates into a more complex Arctic security context. Retreating sea ice has opened parts of the Arctic Ocean to seasonal access, resulting in increasing civil and military activity and enhanced demand for presence and response capabilities.[15] Technological progress compresses distance and time and accelerates the pace of conflict, constricting decision space. General Gregory M. Guillot, commander of USNORTHCOM and North American Aerospace Defense Command (NORAD) has stressed that advanced threats are increasingly difficult to detect and can strike with limited warning, reducing the time and response options available to national leaders.[16]

The Kremlin's Challenge in the Arctic

The Office of the Director of National Intelligence's (ODNI) 2026 threat assessment, in line with previous US strategy documents and allied assessments, highlights that Russia is the

“primary challenge in the Arctic as it aims to further its interests in the region as part of broader global balance-of-power competition”.[17] Since 2012, Moscow has reprioritized investment in modernizing its Cold War military bases[18] and logistics nodes, new weapon systems (many of which are tested in the Arctic), and dual-use platforms. Since 2022, Russia’s defense-industrial base has been on a wartime footing, expanding output of munitions, missiles, and other systems to sustain prolonged conflict and regenerate combat power over time. This buildup has been enabled in part by continued Chinese supply of critical dual-use components, machine tools, and electronics.[19] ODNI notes that Russia invests in ”advanced systems, including counterspace weapons, hypersonic missiles, and undersea capabilities” and ”novel nuclear weapons platforms to supplement its already formidable nuclear air, land, and sea-based triad, complicating US nuclear deterrence calculus".[20] While Russia’s focus on operations against Ukraine since 2022 has impacted the Russian force footprint in the Arctic (primarily in the land domain), analysts assess that Russia will increase its strategic investment and posture in the region if the conflict in Ukraine is resolved or reaches a stalemate.[21]

Dating back to the Cold War, Russia’s Arctic posture has centered on protecting its “bastion” and strategic nuclear forces, including its sea-based second-strike capability based around the Kola Peninsula. That posture also extends control-and-denial pressure outward to impose costs on western access and complicate operations across North Atlantic approaches.[22] As NATO Supreme Allied Commander Europe (SACEUR) Alexus G. Grynkewich highlights, this threat is especially acute in the Greenland, Iceland, United Kingdom-Norwegian (GIUK-N) Gap, a strategic chokepoint and a vital detection area for any potential Russian nuclear-powered and guided missile submarines before they can reach cruise-missile range of the US east coast.[23]

Russia uses calibrated conventional military multi-domain operations and subthreshold activity (often paired with nuclear coercion) to contest access, complicate allied decision-making, and test allied responsiveness.[24] The Kremlin’s hybrid playbook targets political, operational, legal, and societal seams to exploit ambiguity and challenge attribution, thereby enabling limited, deniable actions, creating strategic dilemmas slowing allied responses, and shaping the battlespace before and during crises.[25] Hybrid activities include monitoring and/or suspected targeting of critical infrastructure nodes, illicit maritime-based activity drawing on assets of the “shadow fleet”, electronic warfare (impacting aviation and maritime security, most pronounced in recent years in the Baltic Sea and Northern Norway and Finland), coupled with influence operations.[26]

While the United States treats the Atlantic and Pacific theaters as regionally distinct, Moscow views the Arctic Zone of the Russian Federation (AZRF) from the Barents Sea to the Bering Strait as a single, connected theater.[27] Following Finland’s and Sweden’s NATO accession, Russia, moreover, updated its military posture[28] and C2 to enable greater strategic continuity between the Arctic and Baltic theaters. This logic reinforces the need for allied planning to avoid compartmentalizing Arctic activities by separating them into subregional problem sets—a tendency that adversaries can exploit.

China's Challenge in the Arctic and Beyond

Even as Russia remains a serious and persistent challenge to US and transatlantic security, Washington's priorities have increasingly shifted toward China, which is broadly seen as the most comprehensive and robust military threat to US national security in the medium and long term. China is working to close technological, economic, and military gaps with the United States by modernizing cross-domain capabilities (including long-range missiles) to deter the United States and support Beijing's expanding global ambitions.[29]

China's Arctic engagement is also growing. Its scientific, commercial, and civil activities in the Arctic support dual-use purposes including the development, testing, and operationalization of polar capabilities from icebreakers to undersea technologies – in line with its military-civil fusion strategy (MCF).[30] The growing awareness of China's military ambitions, economic and strategic interest in the polar regions, and the dual-use potential of infrastructure investments and scientific exploration in the Arctic has severely limited Beijing's cooperative and investment opportunities in the seven Arctic states that are also NATO members.

Meanwhile, China has increasingly looked to Russia and the "global commons", to enhance its Arctic access.[31] Beijing has invested in Arctic-relevant space infrastructure, including optical imagery satellites, Synthetic Aperture Radar (SAR), and satellites supporting positioning, navigation, and timing (PNT).[32] These kinds of assets can support situational awareness, environmental monitoring, navigation, and military operations. Regular scientific missions of its (gradually) expanding icebreaker fleet support seabed mapping and the exploration of the undersea environment, including its acoustics, paired with relevant capability development, technology testing, and the expansion of dual-use scientific and commercial presence.[33] These developments create enabling conditions that could support future military or hybrid activity[34] and strengthen China's position to support long-term competition with the United States.

Russia–China Cooperation

Russia's and China's cooperation in the Arctic has evolved significantly within the last decade. While their partnership has grown across the economic, scientific, civil, and military level, analysts assess that significant limits remain in the strategically sensitive Arctic, where Russia has historically acted as a "gatekeeper".[35]

Nonetheless, they have strengthened coast guard and military cooperation in the Arctic since 2023.[36] Recent incidents involving Russian and Sino-Russian long-range aviation activity near Alaska and joint coast guard patrols in the Bering Sea, requiring responses by NORAD and the US Coast Guard,[37] illustrate that signaling in the North Pacific can translate directly into Arctic operating requirements for the United States and underscore cross-theater linkages.

The 2026 US National Defense Strategy (NDS) explicitly identifies a "simultaneity problem", warning that "one or more potential opponents might act together in a coordinated or opportunistic fashion across multiple theaters."[38] The most demanding challenge for US and allied planners arises from the interplay between Russia's and China's activities. Their approaches to the Arctic are not identical, but they are increasingly complementary and

coordinated. Russia provides geographic access, operational experience, and a standing military presence. China offers industrial capacity, technological development, long-term strategic depth, and a market for Russian energy exports.[39]

These dynamics together increase Russia's and China's ability to oppose the United States and its allies through civil, military, and political activities and to exploit gaps and seams. This includes the use of tools under the threshold of armed conflict, which complicate attribution, response, and prioritization. Russia has long focused on exploiting political and military vulnerabilities, especially where legal authorities, operational ownership, and decision processes do not align with how risk converges across the Arctic-North Atlantic-North Pacific space. Beyond China's "lawfare"[40], Beijing appears to be drawing lessons from Russian operations in Ukraine and a security environment increasingly defined by asymmetric threats.[41] Despite nuanced distinctions, Russian and Chinese military thinking emphasizes disabling critical infrastructure, networks, and key logistics and operational nodes to disrupt an opponent's ability to coordinate and respond in crisis or conflict.[42]

Gaps and Seams in Multi-Domain and Multi-Actor Operations

Distance, harsh environmental conditions, and the demanding operating environment in the Arctic underpin preexisting gaps in monitoring, communications, power generation, and dual-use infrastructure. These physical shortfalls are closely linked to limited intelligence, surveillance, and reconnaissance (ISR) and communications capabilities, both of which are foundational to sustaining a common operating picture and seamless C2. They are further exacerbated by limited force projection capacity, especially cold-weather forces, which require specialized training and polar-capable equipment.

Because Arctic operations demand highly specialized, niche capabilities and supporting infrastructure—both of which are often costly and available only at limited scale, filling gaps is challenging. Resource demands compete with other regional and functional priorities, constraining investment in Arctic-specific assets. These gaps limit operational options in the region and heighten interdependence among allies,[43] constraints that can be aggravated when scarce forces and enabling capabilities are diverted to other theaters under competing demands. The operational effect of gaps is often less about the absolute absence of capability than it is about the difficulty of generating enough relevant capacity, at the right time, across multiple theaters.

Seams at the intersections of systems can be geographic or functional. Regional seams are most acute where operational problems span more than one AOR or jurisdiction, but planning and decision authorities remain partitioned. Functional seams may emerge from mission and governance boundaries, including at the intersection of interagency, civil-military, and public-private responsibilities. Civil authorities, coast guards, border services, and defense forces may hold different information, thresholds, and authorities across allied governments.[44] These seams matter because Arctic posture, mobility, and resilience depend heavily on dual-use infrastructure and commercial logistics. Across the full-spectrum of conflict, unclear

responsibilities, differing priorities, and information-sharing constraints can slow the transition from routine awareness to coordinated response.[45]

Seams also emerge in the integration of data and decision-making systems, whose full efficacy depends on fusing, processing or disseminating data. US modernization efforts explicitly aim to optimize this process to support situational awareness, accelerate timelines, and achieve decision advantage utilizing AI.[46] Project Maven (formally established in 2017) was created to accelerate integration of big data and machine learning and to augment or automate processing, exploitation, and dissemination (PED) for ISR workflows.[47] NATO and individual allies are pursuing parallel objectives along similar lines.[48] Commercial actors are playing an ever more important role[49] in ISR, which requires strategic and targeted private-sector engagement. Arctic awareness and resilience increasingly depend on commercially provided data and services, including synthetic aperture radar (SAR) imagery, satellite communications, and cloud-scale analytics, which can reduce latency but also introduce governance, contracting, and security seams that must be addressed.[50]

Functional seams emerge when cross-domain effects are required, but authorities, data, and processes remain organized in stovepipes. They are especially significant in a context in which both the United States and NATO are focused on enhancing multi-domain integration.[51] Across all these frameworks, the decisive variable is often the ability to synchronize sensing, C2, and effects across all domains at the speed of relevance—and to do so with allies.

Seams become more consequential as the United States and key allies field more integrated next-generation systems, and as stealth, autonomy, and long-range precision capabilities interact with the realities of contested and degraded connectivity. Under conditions in which platforms must limit communications, operational success relies more heavily on disciplined pre-crisis planning, common data standards, rehearsed handoffs, and the ability to share information at the appropriate classification and releasability level early enough to shape joint and combined options.[52]

Gaps and seams become increasingly exploitable under conditions of cross-theater simultaneity and in the context of resource and force-deployment prioritization. Adversaries can use probing activity and ambiguous, below-threshold actions to force time-consuming coordination across areas of responsibility (AORs), jurisdictions, and rules of engagement as they seek to fragment the operational picture and slow and hamper consultation, resulting in a delayed or incoherent response.

The Evolution of US and Allied Arctic-Relevant Command Structure

Arctic-relevant command responsibilities across the different Arctic regions are multifaceted, at times distinct, but generally interdependent. They vary by geography and mission. Higher population and infrastructure density in the warmer European Arctic enable comparatively more robust military mobility, while harsher conditions in the North American Arctic impair

mobility, communications, and basing infrastructure. These differences generate distinct operational challenges and mission priorities across subregions.

In the North American Arctic, the command architecture centered on NORAD and USNORTHCOM is focused primarily on aerospace warning and control and missile defense. This includes the detection of and warning against threats approaching Alaska, the continental United States, Canada, and Greenland, to support missile defense via sea-based and land-based interceptors in Alaska and in the contiguous United States.[53] The increased frequency and scale of naval activity around Alaska and in the Bering Sea tie Arctic warning and response timelines to Indo-Pacific dynamics and growing Sino-Russian activity in the North Pacific.

US situational awareness and escalation management, and the capacity to defend against Russia, depend heavily on allied posture, basing, and activity[54] in the European Arctic and High North. Norway and other Nordic allies, as well as the United Kingdom (UK), form a forward layer of maritime and airspace surveillance to monitor and, if needed, deny Russian activity and power projection into the North Atlantic via the GIUK-N gap, as well as toward North America.[55] Meanwhile, US enablers, forces, logistics, and sustainment remain critical for the defense of NATO's northeastern flank.

Finland's accession to NATO has, moreover, extended the Alliance's border with Russia by roughly 830 miles, demanding a more pronounced NATO focus on the land domain in Northern Europe, in addition to evolving air and maritime dynamics.[56] This underscores that Arctic defense requires the coordination of complex multinational, multi-domain operations across diverse, strategically interconnected theaters. These interconnections elevate the importance of integrated planning across US COCOMs, NORAD, the US Coast Guard, and NATO (and coordination with Indo-Pacific partners) so that indications and warning (I&W), authorities, and response options are aligned in advance and seams between regions do not become gaps in crisis response.

Assessing the Evolving US Arctic Command Structure

US COCOM boundaries, and thus Arctic seams, are drawn through the Unified Command Plan (UCP), which is reassessed every two years by the chairman of the Joint Chiefs of Staff. As the strategic environment, the threat landscape, and technologies evolve, the Department of War's (DoW) global map outlined via the UCP[57] is adapted accordingly, with respective responsibilities divided up among several geographic and functional COCOMs.[58] The deliberation process and the eventual operationalization of these changes is often lengthy.[59]

Over the last two decades, the US national security apparatus has been adapting its concept and capability-development to fit a more competitive environment, balancing centralization and decision speed with specialized and regionalized needs. Rapid advances in AI, autonomy, microelectronics, space, cyber capabilities, and quantum sciences are compressing decision cycles.[60]

US responses have combined concept-driven modernization with institutional adaptation. In recent years, the US military has updated its approach to ensure that its doctrine and capabilities meet operational and theater requirements, including the development of a new Joint Warfighting Concept, updated in 2023.[61]

In this context, the ultimate challenge within US COCOMs is to ensure that, as priorities shift, the United States retains an agile and available force posture, situational awareness, and C2 to effectively deter and respond to multi-theater contingencies that involve the Arctic.

US Arctic Command Responsibilities

Although their AOR stretches beyond the Arctic, three geographic US COCOMs hold intersecting, but distinct Arctic responsibilities, which shape their strategic lens and priorities.

- **USNORTHCOM,** headquartered in Colorado Springs, Colorado, focuses on homeland defense, including threats emerging from the North. Its AOR spans the continental United States, Alaska, Canada, Mexico, and parts of the Caribbean.[62] Through the Arctic lens, its AOR encompasses Alaska and Canada, and following a UCP adjustment, Greenland, which was reassigned from US European Command (USEUCOM) to USNORTHCOM.[63]
- **USEUCOM**, headquartered in Stuttgart, Germany, whose Commander also serves as NATO's SACEUR, focuses on "safeguarding the homeland while enhancing security across the Euro-Atlantic region"[64] and holds relevant responsibilities for the defense of the High North. Its AOR covers Europe, Russia, the Caucasus, and Türkiye. Its Arctic-related AOR covers the European High North and a part of North Atlantic.
- **USINDOPACOM,** headquartered on the island of Oahu, Hawaii, focuses on cross-domain operations that "defend the homeland, deter strategic attack, counter aggression, protect US interests throughout the Indo-Pacific". Its vast AOR stretches from the US west coast to India, and from Antarctica to the North Pole. As a result, its Arctic-related AOR covers the North Pacific, Bering Strait, to the North Pole.[65]

They are supported by globally tasked enablers, including functional COCOMS: Cyber Command (USCYBERCOM), Special Operations Command (USSOCOM), Strategic Command (USSTRATCOM) and Transportation Command (USTRANSCOM) and their corresponding service component commands[66], as well as by cross-cutting, enabling space capabilities provided by US Space Command (USSPACECOM) that underpin multi-domain operations.

Regionally, these functional components are especially critical for space-enabled services (PNT, communications, missile warning), cyber defense of C2, and strategic deterrence architectures that operate over polar trajectories. For instance, USSTRATCOM coordinates closely with USNORTHCOM on "assessing strategic missile threats and prioritizing locations for defense against attacks by nuclear-armed adversaries".[67] However, these functional dependencies are where capability gaps and organizational seams most often overlap in practice since regional COCOMs may compete over functional support, especially if resources

are stretched. “Blind spots” may also appear where interdependent cross-domain effects (e.g., cyber and space) require closely coordinated planning.[68]

Similarly, while no COCOM “owns” forces, US posture in the North American Arctic depends on the availability and readiness of well-equipped and Arctic-trained personnel. Such forces are limited. Moreover, they may support several mission sets and regional requirements. This is the case for forces under Alaska Command (ALCOM), which serves as the joint headquarters responsible for Arctic-relevant operations in Alaska. While ALCOM, which is headquartered at Joint Base Elmendorf-Richardson (JBER) in Anchorage, is USNORTHCOM’s subordinate unified command, key elements within this structure, such as the 2022-reactivated US Army’s premier extreme cold-weather force—the 11th Airborne Division, also known as “Arctic Angels—retain dual responsibilities spanning both Arctic and Indo-Pacific missions.[69] This arrangement, with dual-reporting lines to USINDOPACOM and USNORTHCOM, keeps the division flexible; However, it also limits operational specialization given limited training opportunities in highly complex and diverse environments and highlights a potential challenge in a dual-contingency crisis with demands in the Arctic and Indo-Pacific. In this context, close cross-COCOM coordination between USNORTHCOM and USINDOPACOM remains essential in bridging operational Arctic seams, as reflected in joint exercises such as Northern Edge, which strive to enhance integration.[70]

The United States has made efforts to expand regional personnel resources across different services. A noteworthy development in the North American Arctic context is the expansion of US Marine Corps presence in Alaska through “Campaign Alaska”. Specifically, the Marine Rotational Force—Alaska will "conduct persistent, multi-domain expeditionary training and experimentation in Alaska", strengthening homeland defense, interoperability with allied and joint forces, and boosting Fleet Marine Force's ability to work in Arctic conditions.[71] Concurrently, the permanent Supporting Arms Liaison Team—Alaska "will be based out of Joint Force Elmendorf-Richardson by fiscal year 2027 and include service members from the 6th Air Naval Gunfire Liaison Company. It will also work with joint forces, allies and the community."[72]

Mirroring USNORTHCOM’s AOR, Special Operations Command North (SOCNORTH) also plays an important role; It supports homeland defense by deterring and combating irregular threats, and facilitating cooperative action with mission partners including in Arctic contexts.[73] Aligned with USEUCOM’s AOR, Special Operations Command Europe (SOCEUR)[74] supports US homeland defense from the European continent by strengthening deterrence and European security collective capabilities and interoperability; in the latter case SOCEUR forces practice rapid deployment and unified operations in Arctic and sub-Arctic regions through the US-led ADAMANT SERPENT exercise.[75]

Responsibilities relevant to Arctic security and presence are distributed across the DoW and other federal actors, including the US Coast Guard (USCG) whose statutory missions span maritime safety, security, law enforcement, and environmental stewardship.[76] USCG plays a vital role in safeguarding US Arctic interests and security via the USCG Atlantic Area

Command and USCG Pacific Area Command. The latter also includes the Arctic district headquartered in Juneau, Alaska. USCG closely cooperates with USNORTHCOM and regularly contributes to bi- and multi-national exercises spanning the Arctic.[77] Recognizing their respective seams, in 2023 the USCG established the Task Force Arctic to synchronize its operations in the theater.[78] Moreover, to protect North American maritime approaches, the US and Canadian navies and coast guards strengthened coordination through the Atlantic Maritime Security Coalition, improving interoperability, joint operations, and Arctic and North Atlantic maritime security.[79]

Coordination between USNORTHCOM and USEUCOM is essential given that their AORs intersect in the strategically important GIUK gap—an area in which the ability to monitor, deny, and control the maritime space would be critically important in a contingency. This command relationship has grown in importance following the reallocation of Greenland under USNORTHCOM in June 2025[80], further discussed in the following chapter.

In this context, US Navy component commands[81] further illustrate important links between the US and NATO. The reactivation of the US Second Fleet in 2018[82] strengthened the ability to project forces from North America into the North Atlantic and Arctic.[83] Its current commander's dual role as commander of NATO's Joint Force Command Norfolk (JFC Norfolk) reinforces transatlantic C2 linkages. Similarly, from the European continent, the US Sixth Fleet covers about half of the Atlantic from the Arctic to the Antarctic, adjacent European and Eurasian seas, all of Europe and Russia, and a substantial portion of Africa.[84] Its commander also heads NATO's Naval Striking and Support Forces STRIKFORNATO, ensuring seamless integration of US carrier and expeditionary strike groups into NATO operations as forces transition to Alliance command.[85] The NATO section will explore how shifts in US NATO leadership roles will affect these dynamics.

North American Integrated Air and Missile Defense

The bi-national command of NORAD has served as the backbone of North American aerospace warning and control, including detection, tracking, and assessment of air and missile threats since its founding in 1957. USNORTHCOM's commander is concurrently the commander of NORAD, and alongside a Canadian deputy commander, facilitates robust and institutionalized cooperation that secures North American maritime, air, and space approaches. [86] The commands' co-location at Peterson Space Force Base in Colorado Springs and their integrated binational command structure with additional regional headquarters in Alaska and Canada allow for seamless communications to support a single warning and control mission.

The US Space-Force-operated Pituffik Space Base in Greenland, which is not part of NORAD but is within USNORTHCOM's AOR, contributes to this wider early-warning architecture through its missile-warning, missile-defense, and space-surveillance functions.[87] Together, NORAD and USNORTHCOM integrate data from these systems into a shared operational

picture that supports time-sensitive decision-making for homeland air and ballistic-missile defense.[88]

NORAD's maritime warning mission established in 2006 is a bi-national command responsibility shared between the United States and Canada. On the US side this responsibility falls under USNORTHCOM**,** executed by the US Navy's Fleet Forces Command (acting as US Naval Forces Northern Command). On the Canadian side it falls under Canadian Joint Operations Command (CJOC), which oversees the seamless integration of the Canadian Armed Forces to execute this mission.[89]

As air and missile threats—and interconnected cyber and Electronic Warfare (EW) challenges—evolve, resulting in a more complex security environment and compressed warning timelines, the United States and Canada are investing in joint and parallel upgrades into their integrated situational awareness, warning, and C2 architecture.

Current NORAD modernization projects, including major investments in over-the-horizon radar and related surveillance infrastructure, reflect efforts to update the legacy warning architecture that was designed to defend against Cold War-era threats (ICBMs). Updates reflect the need to keep pace with technological progress and defend against cruise missiles and other non-ballistic aerospace threats in an increasingly contested Arctic environment in the era of great power competition.[90]

The United States is also pursuing longer-term modernization initiatives intended to augment today's architecture. These include the directive to develop a future-oriented, multilayered, next-generation homeland missile-defense "system of systems" under the program known as "Golden Dome for America".[91] Concurrently, US Space Force polar satellite communications systems, including the Enhanced Polar System–Recapitalization (EPS-R) project, provide secure connectivity and C2 for high-latitude operations[92]—including in the scope of the cooperative US-Norwegian Arctic Satellite Broadband Mission.[93] These systems support a shared operational picture that enables time-sensitive decision-making for homeland air and ballistic missile defense.

Taken together, Arctic defense today is less about assigning ownership to a single command than about ensuring that cross-command, cross-service, and multinational linkages are fast, practiced, and resilient—especially at the USNORTHCOM–USEUCOM boundary and at the NORAD/USNORTHCOM–NATO interface. The following chapter will discuss NATO's evolving role in the North Atlantic and Arctic in response to a changing threat landscape to analyze seams that need special attention.

NATO's Evolution and Arctic Implications

After the post-Cold War shift to expeditionary crisis management, Russia's illegal annexation of Crimea in 2014 and, more decisively, its full-scale invasion of Ukraine in 2022 compelled the NATO alliance to re-center on its core mission of collective territorial defense. In response to the changed threat environment, Finland and Sweden joined NATO in 2023 and 2024 respectively.[94] This development bolstered cohesion, defense capacity (especially in the land domain), and logistics links along NATO's northeastern flank and changed security dynamics across the Baltic and Arctic nexus. Russia's war against Ukraine also initiated an unprecedented increase in defense spending across allied capitals (in tandem with support and security assistance for Ukraine).[95] In this context, NATO transitioned for the first time since the Cold War to a system of standing plans, clearly assigned roles, and reinforcement pathways—de facto reducing seams under compressed warning time. US leadership has been critical in this process.

NATO's adaptation and rapid transformation now intersect with a debate driven by the current Trump administration about what a more "reciprocal" alliance—often captured in shorthand as "NATO 3.0", coined by Under Secretary of War for Policy Elbridge Colby—should look like.[96] The concept envisages a "more balanced, effective, and resilient Alliance" that leverages each side's "respective strength". In essence, it argues that Europe should become the primary provider of conventional deterrence and defense in Europe, while the United States should continue to provide extended nuclear deterrence, but focus more on managing competing theater demands. Separating these interconnected missions is challenging in practice and may undermine deterrence.

European capitals are making meaningful progress toward the goal of enhancing their defense capacity and leadership in NATO. However, the sustainability of this shift—including defense plans and capability targets—depends on adequate planning horizons and timely allied resource alignment. The Hague Declaration commits allies to invest 5% of GDP[97] annually by 2035, structured as at least 3.5% for core defense requirements and up to 1.5% for defense-and security-related spending (including resilience, critical infrastructure, networks, innovation, and the defense industrial base), supported by annual plans and a review in 2029.[98]

While politically significant[99], these investments are inherently lagged; budgets translate gradually into deployable forces, stockpiles, integrated air and missile defense, logistics capacity, and trained personnel at scale. If shifts in US signaling and leadership transitions outpace Europe's ability to convert investment into usable capacity and capability to secure the European theater, unintentional gaps may appear and be exploited by adversaries.

NATO Command Structure and Arctic Defense Since 2022

NATO's military command structure provides the primary framework through which allied deterrence and defense in the European Arctic and North Atlantic are planned and executed. It

is further bolstered by closely connected bilateral[100] and minilateral formats, including the UK-led Joint Expeditionary Force[101], which offer more flexible response options.

At the strategic level, NATO's military command structure is headed by two strategic commands:

- Allied Command Operations (ACO) at Supreme Headquarters Allied Powers Europe (SHAPE) in Mons, Belgium, and
- Allied Command Transformation (ACT) in Norfolk, Virginia.

Operational control is exercised through Joint Force Commands, supported by tactical domain-specific component commands.[102]

NATO's regional defense plans are coordinated and operationalized by three Joint Force Commands (JFC Brunssum, JFC Naples, and JFC Norfolk) with distinct AORs. While JFC Brunssum is responsible for much of northern/eastern Europe,[103] JFC Norfolk, as the newest and only JFC on US soil, is the key operational hub for the North Atlantic and High North, linking maritime control, transatlantic reinforcement, and defense of the Alliance's northern flank.[104] At the 2023 Vilnius Summit[105], NATO approved a new generation of defense plans, including "Regional Plan Northwest", which spans the High North and the North Atlantic, led by JFC Norfolk. Following adjustments finalized in December, JFC Norfolk now includes all Nordic allies—Finland, Iceland, the Kingdom of Denmark, Norway, Sweden, and the United Kingdom—under one command, with an AOR spanning from Florida to Finnmark. This is a significant change and one that creates coherence in planning between Northern Europe and North America.[106]

JFC Norfolk leads Arctic Sentry, launched in February 2026[107], a NATO-wide multi-domain enhanced Vigilance Activity (eVA) that synchronizes allied military activities in the Arctic and High North under a NATO umbrella. Unlike NATO's preceding eVAs—Baltic Sentry and Eastern Sentry, both of which were launched in direct response to concrete Russian actions including airspace violations and undersea infrastructure threats—Arctic Sentry reflects NATO's effort to get ahead of emerging risks in a theater where national forces have historically operated in parallel or in smaller formations rather than under a set NATO framework. Arctic Sentry aligns existing national, bi-national, and multi-national activities. It reflects NATO's broader shift toward day-to-day deterrence—emphasizing presence, monitoring, and integration over episodic deployments or crisis-driven reinforcement cycles.

NATO's component commands[108]—Allied Maritime Command (MARCOM), Allied Air Command (AIRCOM), and Allied Land Command (LANDCOM)—directly report to ACO while providing tactical support functions to JFCs for exercises and missions. MARCOM[109], headquartered in Northwood, UK, exercises operational command over NATO's four Standing Naval Forces providing continuous maritime presence, readiness, and situational awareness across the North Atlantic, Baltic Sea, Mediterranean, and Black Sea. In the heightened security environment, MARCOM has increasingly focused on anti-submarine warfare, the protection of transatlantic and regional sea lines of communication, and the security of critical undersea

infrastructure, particularly in the North Atlantic, High North, and Arctic approaches. AIRCOM, headquartered in Ramstein, Germany, oversees air policing and integrated air and missile defense and coordinates space-enabled support through the NATO Space Operations Centre. LANDCOM, situated in İzmir, Türkiye, ensures that land forces—including increasingly capable Nordic forces—are integrated into NATO defense planning and exercises.

An important Arctic-relevant institutional development within the air command architecture is the establishment of the Combined Air Operations Centre (CAOC) in Bodø, Norway, which became operational in 2025[110] and is in the process of being staffed up. CAOC Bodø serves as NATO's third permanent CAOC[111] in Europe, alongside Uedem and Torrejón, and significantly strengthens allied air command and control in the High North. Its location north of the Arctic Circle enables rapid reaction, improves regional situational awareness, and facilitates direct integration with Nordic national air forces following Finland's and Sweden's accession.

CAOC Bodø supports national Norwegian and NATO operations and is responsible for planning, directing, and coordinating allied air operations across the Nordic region, the North Atlantic approaches, and Arctic-adjacent airspace. It also assumes responsibility for Norway's long-standing Quick Reaction Alert (QRA) mission[112], anchoring a continuous air-policing posture that Norway has maintained for decades. Beyond geography, the Bodø CAOC adds redundancy and resilience to NATO's air C2 architecture and is closely linked to JFC Norfolk's joint operational role, reinforcing the alliance's ability to conduct integrated, multi-domain operations in a contested Arctic environment.

These command-level adaptations are reinforced by Finland's and Sweden's hosting of new NATO bases and structures that are in the process of being stood up or scaled up. This includes Forward Land Forces (FLF) Finland, anchored in Rovaniemi with dispersed support elements. FLF Finland is expected to be operational by summer 2026, with Sweden serving as the Framework Nation and including multinational forces from Denmark, Finland, France, Iceland, Italy, Norway, and the UK.[113] These arrangements, together with investments in logistics corridors in Northern Europe, enhance NATO's ability to train, operate, and reinforce its northern flank—with a strengthened focus on the land domain—in close coordination with national forces.

NATO Command Transitions, JFC Norfolk, MARCOM, and the Evolution of US Engagement

Recently announced adjustments to NATO's command leadership[114] reflect US efforts to shift responsibility to European allies while retaining strategic enablers that are essential for alliance deterrence, escalation management, and transatlantic reinforcement. Concurrently, this shift aligns with broader US efforts to reduce US personnel in Europe. In line with allied preferences, the position of SACEUR, NATO's highest operational commander, will continue to be filled by an American officer, preserving the strategic and historic links between the US (via USEUCOM) and NATO command structures. This arrangement underpins US extended deterrence in Europe. As John Deni and Mark Webber point out, "SACEUR also plays a central

role in NATO nuclear-sharing arrangements, authorised by the White House—which owns the nuclear weapons—and NATO's Nuclear Planning Group to deploy dual-capable aircraft in a crisis."[115]

From a macro level, under announced changes, JFCs will be led by European (typically 4-star) commanders; while the (tactical) component commands will be headed by American (typically 3-star) commanders, with the latter usually accompanied by a smaller staff.

JFC Norfolk's leadership will transition from a US commander who simultaneously commands the US Second Fleet, to a British commander. Until now, a British vice admiral has served as deputy commander—a role that will likely be replaced by a US deputy. This change reflects a deliberate effort to increase European ownership of regional defense execution. This transition does not diminish JFC Norfolk's functional importance; rather, it reinforces its role as a coordinating hub linking North America, Europe, and the maritime domain. Considering the importance of the US Second Fleet headquartered in Norfolk for the defense of the North Atlantic, continued coordination will remain central and will be eased by the co-location of the two commands.

When MARCOM leadership transitions to a US commander, all tactical NATO commands will be headed by US officers. The US assumption of leadership fits squarely within this logic. Maritime control, secure sea lines of communication, and undersea infrastructure protection are foundational to any credible deterrence and defense posture in Europe and the Arctic. By retaining leadership of MARCOM, the United States preserves direct control over the domain most critical for reinforcement and escalation dynamics, even as European allies assume greater responsibility for regional and JFC roles.[116] In line with historical precedent and US ambition to reduce personnel, the MARCOM Commander will likely also command the 6th fleet, meaning they would divide their responsibilities between a NATO and a US command in different regions.

Views on the benefits and downsides of dual commands are divided. While such arrangements may result in greater synergies and quicker coordination between the United States and NATO in crises, they may also limit bandwidth and connections with local forces and staff—especially when the commander is splitting time between two locations. Challenging seams could arise in contingencies that might draw on US component commands and limit the availability of US assets and forces integral to the NATO plans overseen by the regional commands. Moreover, the current relationship between European-led regional and US-led component commands should be retained to ensure smooth operations, and burden sharing. Any tendency to circumvent Europeans in planning and decision-making would be highly problematic and undermine unity of effort.

Key Arctic Gaps and Seams Between the US and NATO Structures

This section highlights the most consequential Arctic-relevant gaps and seams shaping US and allied ability to deter and, if required, defend under conditions of multi-theater pressure. It

focuses on recurring junctions that consistently produce delay or uncertainty, especially at the US–NATO interface and at the boundaries between national or multinational geographic and functional commands.

Arctic Domain Awareness and Intelligence-Sharing

A persistent and foundational gap in Arctic defense is the absence of timely, shared domain awareness and fused I&W. This reflects not only the physical challenges of sensing and persistence across vast distances and unique conditions (resulting in situational awareness gaps), but also limited capacity to rapidly integrate multi-domain inputs into a common operational picture and disseminate that picture across national and alliance systems. These challenges are compounded by longstanding intelligence-sharing and releasability frictions stemming from cultural and technical factors, which impact the time required to move from national collection and analysis to a coalition-usable picture. Both cultural and technical factors, which disincentivize or slow sharing, further exacerbate this. While intelligence cycle security is of utmost importance, restrictive default markings (including SECRET/NOFORN)[117], uneven foreign-disclosure pathways, and cross-domain transfer or data-labeling practices can delay or prevent shared I&W and impede accurate intelligence analysis.

The result can be compressed warning time, inconsistent assessments among allies, and heightened escalation risk driven by uncertainty, especially when attention and high-demand enablers are also being pulled toward other theaters. Such seams are particularly consequential in a fast-paced, technology-driven security environment, a dynamic, that General Guillot underscores. He highlights that "Regional conflicts routinely have the potential to expand into broader crises that directly impact U.S. and allied interests" and may generate homeland defense implications.[118]

Operational AOR Seam

As outlined above, Arctic defense spans multiple AORs, requiring close coordination and integrated planning—especially for multi-vector crises that may place competing demands on the US Joint Force and NATO allies.

First, on the US side, officials have been focused on addressing geographic seams between USNORTHCOM, USEUCOM, and USINDOPACOM through cross-COCOM coordination in operations and exercises. Beyond communication and coordination, this requires the reconciliation of diverse mission sets of these commands that define their respective threat perceptions, prioritization, and preferred response options. INDOPACOM's vast AOR and set of responsibilities—of which the Arctic only forms a small subset—often eclipses Arctic considerations.

The reallocation of Greenland under USNORTHCOM—while logical from a geographic and mission-set standpoint—adds a complicating coordination layer. Denmark, which oversees Nuuk's foreign and security policy, continues to fall under USEUCOM. Since USNORTHCOM's and JFC Norfolk's AORs moreover overlap in Greenland, this shift requires further attention as such multifaceted connections may complicate coordination in a contingency that requires

fast decision making. The newly announced "Nordic Bridge" mechanism will enhance collaboration between NORTHCOM, NORAD, EUCOM, and US Special Operations Command Europe. This is a welcome development given the need for coordination. It remains to be seen how it is implemented.[119]

Second, at the NATO level, the interface between JFC Brunssum and JFC Norfolk, linking the Baltic region and Northern Europe to the North Atlantic and High North—often reduced simplistically to a seam across the Baltic Sea—frequently draws attention given the strategic continuity between these theaters. This seam is aggravated by Russia's capacity to introduce (and instrumentalize fear of) horizontal escalation dynamics via its civil and military activities across both regions, complicating regional responses.

Political Defense Commitments and the Operational AOR Seam

Another noteworthy disconnect lies between collective-defense commitments covering NATO territory and operational NATO JFC AOR boundaries, which do not always equate. Specifically, while Alaska and Canada fall under NATO's collective defense provision, NATO operational command arrangements for the North Atlantic and High North do not map cleanly onto the defense of North America. This reflects the expectation and the longstanding understanding that the United States and Canada would respond to contingencies in the area with national forces. However, it is conceivable that some form of allied support could be required under specific circumstances. In the absence of exercised procedures and habitual relationships, this misalignment can create uncertainty during Arctic incidents. Even when responsibilities are clear, insufficient operational experience in a unique and demanding environment can be expected to pose acute challenges. This makes exercising across seams particularly important.

Cross-Command Integration for Global Enablers

As discussed, Arctic-relevant operations overseen by regional commands routinely require globally allocated enablers (strategic mobility and sustainment, ISR, communications and space-enabled services, cyber effects, and special operations), none of which are owned by the regional commands. On the US side, these are captured through the four functional COCOMs of USCYBERCOM, USSOCOM, USSTRATCOM, and USTRANSCOM, as well as USSPACECOM, which is considered a geographic command, but which provides global effects.

On the NATO side, this encompasses the tactical Component Commands (AIRCOM, MARCOM, and LANDCOM) and additional tactical C2 elements. In this context, competition between commands may result in gaps of forces and appear when supported–supporting relationships are unclear, prioritization across theaters is contested, or authorities are not pre-aligned for cross-AOR execution under simultaneity.

Misalignments in threat perceptions and the operating picture can reinforce planning misconceptions about the availability of enablers and platforms. In an Arctic context, the limited availability of polar-capable platforms and forces can result in unexpected wait times

until relevant assets reach their destination. These slow deployment processes and limited regional posture do not match the requirements of a fast-paced threat landscape.

Human and Organizational Connectivity Risks

Finally, many critical processes are sustained less by formal structure than by human and organizational dependencies. Day-to-day integration and information- and intelligence-sharing often hinge on personnel across headquarters, the depth and empowerment of liaison networks, and informal knowledge of processes and counterparts.

Where personnel reductions, turnover, or thin liaison coverage intersect with newly created or expanding command interfaces, routine coordination can degrade quickly. In the Arctic, these challenges become especially acute at the interface between NORAD/USNORTHCOM and JFC Norfolk, where homeland defense, the GIUK–Norwegian Sea corridor, and transatlantic reinforcement timelines converge.

These risks are magnified during periods of transition, as NATO and national command arrangements evolve through leadership changes, staffing adjustments, and the operationalization of new headquarters or missions. Deni and Webber argue that US leadership reductions in NATO not only affect force and command distribution, but are expected to have a qualitative impact, since the US presence "has a cohering effect, facilitating the dissemination of American military concepts and doctrine among the allies".[120]

Unless connectivity, procedures, and institutional memory are deliberately maintained and exercised, transition periods can widen seams and impede interoperability precisely when Arctic-relevant contingencies—characterized by short warning times and intrinsic cross-command coordination requirements—demand maximum clarity and coordination.

Recommendations

The following recommendations focus on key lines of effort (LoE) to reduce exploitable seams between US and allied command arrangements in the Arctic. Collectively, they emphasize repeatable processes, clarity on roles and resourcing, and practical measures to improve decision speed while protecting sensitive information from adversary exploitation.

LoE 1: Bolster Clarity and Coordination to Maintain Credible Deterrence and Defense

1. Provide clarity about US support to enable realistic planning

As Arctic and NATO allies assume greater responsibility for regional defense, the United States should communicate both its expectations and its commitments to provide critical reinforcements and enablers in contingencies through secure channels, with sufficient clarity and planning horizons to enable realistic national and NATO-wide planning. This is the foundation necessary for European allies to take greater responsibility while sustaining a credible posture, to deter and defend against threats coming from or through the Arctic. As

some functions shift and rotate among allies, transition planning and rehearsal should be treated as operational requirements, not administrative afterthoughts.

Proactive consultations with key allies are central to ensuring quick and unified decision-making in peace, crisis, and conflict, ultimately saving time and resources. For Arctic contingencies, consultations should include or be informed by key stakeholders from relevant US and allied commands. Engaging these stakeholders ensures that planning reflects the full range of capabilities and constraints that bear on a contingency plan and helps build a shared understanding of how responsibilities transfer across theaters and command boundaries.

Clear guidance should be matched by predictable information flow. Unity of purpose in the Arctic depends less on episodic exchanges than on habitual, routinized coordination across the strategic, operational, and staff level, that normalizes shared analysis and early warning. Maintaining an information and decision advantage in a contested environment rests on a set of mutually reinforcing foundations. Trust among partners, built slowly through demonstrated competence and discretion, is the precondition for sharing sensitive intelligence and acting on it quickly. Adherence to international law and to common rules of engagement standards gives allies confidence that joint and coalition operations will remain lawful, predictable, and politically defensible. Reciprocity keeps the burdens and benefits of cooperation balanced, while reliable commitments allow allies to plan and act with confidence.

2. Optimize (Arctic-relevant) intelligence-sharing

Predictable information-sharing also requires revisiting how intelligence is marked and disseminated to ensure that allies operate based on common strategic and operational assumptions.

Technical compatibility is a baseline requirement, but is not in itself sufficient. Culture and established, classified, and tested channels are equally important. Where trust, responsible and secure handling of information, and reciprocity are lacking, detrimental information gaps are bound to arise, leading to potential miscalculations during crisis. Established and frequently tested channels become even more important where technical standards vary and require adjustments.

A targeted review should assess Arctic-relevant data categories most frequently needed for warning and response—such as selected tracks, cues, and cyber indicators—to identify where default controls (including NOFORN) introduce avoidable friction among trusted allies.

Where feasible, the US intelligence community should crystalize and pre-approve tearlines defining access by classification level and foreign disclosure pathways for defined scenarios, in line with current efforts to enhance intelligence-sharing "speed, integration and innovation".[121] This warrants a closer look at enhanced intelligence-sharing channels with key NATO allies. As the United States considers expanding its military footprint in Greenland under the auspices of the bilateral 1951 Defense of Greenland Agreement and Denmark steps up investments in Arctic capabilities and monitoring infrastructure,[122] relevant officials should consider an

enhanced intelligence-sharing relationship between the United States and Denmark and NORAD and Denmark. An enhanced intelligence-sharing format could also be applied to key Arctic allies with critical functions for US homeland defense. This includes Canada, Denmark, Norway, and the United Kingdom. Alternatively, enhanced sharing could be achieved through expanded preauthorized releases.

The goal is not broad liberalization, but a predictable releasability that enables rapid convergence on shared I&W while protecting sources, systems, and methods. Without this step, even well-established coordination channels will remain bottlenecked.

LoE 2: Enhance Links Across NATO-US COCOMS

1. Compensate for reduced US NATO staffing with reinforced allied liaison capacity

Improved information- and intelligence-sharing relies on human mechanisms to work in practice. Where reductions in US billets, including at NATO intelligence fusion nodes,[123] have weakened day-to-day connectivity, increasing reliance on informal networks creates fragility especially during fast-paced, multi-theater contingencies. Seams introduce special risks in contingencies involving the strategically important Arctic, where allied situational awareness and intelligence-sharing are critical to North American and European defense and assessments of escalation dynamics.

Where this connectivity has thinned, seams should be offset by deliberate investment in liaison architecture. This includes adding select allied liaison officer roles at NORAD and USNORTHCOM with explicit responsibility for NATO coordination and releasability facilitation, and ensuring robust, reciprocal liaison links with JFC Norfolk and other relevant NATO commands. Persistent, empowered liaison capacity transforms coordination from a best-effort activity into a reliable function of the system.

This change is especially important in the context of the UCP reorganization that has shifted command responsibility over Greenland from USEUCOM to USNORTHCOM.[124] This reorganization requires closer coordination between USNORTHCOM, USEUCOM, and JFC Norfolk, in addition to bilateral cooperation between the United States and the Kingdom of Denmark (including Greenland).

2. Bolster cross-regional and cross-functional interoperability and planning through exercises

Exercises translate doctrine into capacity and can help clarify cross-seam operations and handover. JFC Norfolk's role in coordinating Arctic exercises and resulting findings under the Arctic Sentry eVA lays the foundation for enhancing interoperability, impact, and lessons learned, while avoiding miscommunication and accidental escalation. With every NATO Arctic state, along with allies such as Germany, the United Kingdom, and France, investing in capabilities integral to safeguarding the High North and adjacent theaters, Arctic Sentry can be viewed as a deterrence-building activity that prepares allies to operate coherently across different AORs in a period of strategic competition viewed as the "new normal". Concurrently,

Arctic Sentry, and other eVAs, demonstrate allied commitment to burden-shifting ambitions.[125] Sustainable investment in this LoE has the potential to evolve and greatly enhance Arctic capacity while bridging seams between the North American/Pacific Arctic and the High North.

NORAD and NORTHCOM exercises such as Arctic Edge, Noble Defender, and Operation Nanook already include key allies and increasingly have sought to bridge seams between regional commands and the civil-military spectrum. Beyond this, regular integration of NATO officers (including JFC Norfolk) into selected NORAD and USNORTHCOM exercises would ensure that procedures, coordination pathways, and personnel relationships are exercised for potential Arctic and North American Article 5 scenarios—before crisis conditions arise.

Arctic exercises (national and multinational) should reflect current planning considerations at the US and NATO level. In today's rapidly evolving environment, planning should be continuously informed and adjusted based on-bottom-up experimentation and rehearsal. Field Training Exercises (FTX) and daily operations need to test and refine regional or domain-specific tactics, techniques, and procedures in units and operational-level nodes under realistic communications constraints. This will allow relevant findings to be integrated into national and multinational plans and operational tempos before a crisis compresses decision time.

In addition, tabletop exercises (TTX) can help assess C2 and component availability in a multi-contingency crisis to identify decision points for prioritization of global enablers and to reduce ambiguity at geographic–functional seams. Where possible and applicable, FTXs and TTXs should also include the private sector to ensure that public-private pathways and limitations during crisis are well understood.

LOE 3: Facilitate situational awareness and technology-enabled decision advantage

1. Invest in shared satellites and sensors to enhance situational awareness

Real-time situational awareness of the operational environment across the High North and the North American Arctic—from seabed to space—is fundamental to effective decision-making across the spectrum of competition and conflict. Yet for many allies, the capabilities required to deliver persistent, all-domain Arctic awareness remain prohibitively expensive. Given that Arctic communications and surveillance require purpose-built and mission-specific satellites set on highly elliptical or polar orbits, this is particularly true in the space domain. Expanding access through co-development of sovereign capabilities, alongside "space-as-a-service" models where appropriate, could make critical technologies more attainable, support interconnected architectures, and help close acute gaps in awareness, warning, and defense.[126]

Commercial options can enable more flexibility and scalability, especially for missions that are less sensitive or that benefit from rapid innovation cycles. This logic extends to elements of the broader missile warning architecture, where commercial capabilities may be able to complement government systems and reduce barriers to access for smaller allies. For missions

that demand higher levels of sovereign control, resilience, and security, however, government-owned architecture will remain essential. In these areas, trusted allies could pool resources to co-invest in nationally controlled or jointly operated systems that would be difficult for any one country to field independently. This principle may also have relevance for the future evolution of Arctic sensing and early-warning networks.

Beyond procurement, collaborative approaches can also strengthen research and development, aggregate demand, and send clearer market signals to industry. In a region where operational requirements are growing faster than many governments' ability to field bespoke solutions on their own, allied cooperation with industry will be essential to building scalable, affordable, and resilient capability.

2. Thoughtfully leverage AI processing tools to complement human judgment

AI-enabled processing tools can help close Arctic-relevant seams in situational awareness and decision speed by triaging large volumes of sensor data and highlighting what merits human attention. In particular, they can add value in image exploitation and target recognition (including pattern-of-life baselining and anomaly detection), rapid screening of commercial and military ISR feeds, automated translation and summarization of time-sensitive reporting, and fusing indicators across domains to flag inconsistencies or emerging risks.

At the same time, Arctic operating conditions create "edge cases" where data can be sparse, noisy, or unevenly labeled, and where context (weather, geography, unusual shipping patterns, and local operational norms) matters for interpretation. In those settings, especially where adversaries may use deception, ambiguity, or deniable activity, AI outputs can miss nuance, hallucinate, overfit to limited training data, or narrow the range of perceived options. The United States and allies should treat AI as an accelerator for analysis and a tool to reduce routine data-processing burdens—not as a substitute for judgment. A human decision-maker must remain responsible for validation, escalation assessment, and final decisions, supported by clear processes for auditing outputs, testing performance in realistic conditions, and ensuring interoperability and releasability across trusted partners. In practice, rather than reducing the staffing needed for resilient C2 in contested environments, this should shift personnel time from manual processing toward higher-order assessment, cross-command coordination, and decision support.

Conclusion

Reform and adaptation are necessary to adjust US and allied command arrangements to a complex and contested security environment that is defined by simultaneity and compressed decision timelines. For US and allied defense planners, the central challenge is to advance both without sacrificing operational coherence and agility. This is especially critical in the Arctic and the North Atlantic, where operational capacity depends on collaboration and where

miscalculation, signaling failures, or delays in allied coordination can carry wider escalation risks.

Structural change should accelerate sound decision-making under realistic conditions: compressed timelines, uncertain attribution, multi-theater demands, degraded communications, austere operating conditions, and the political constraints of coalition warfare. Where national and allied command structures evolve in parallel, coordination is the mechanism by which reform avoids generating detrimental surprises or new friction.

Seams are an enduring feature of complex command arrangements. The operative question is whether they are understood, anticipated, and managed in ways that reflect allied plans, authorities, and decision cycles. Command consolidation can shorten formal chains of command, but it does not remove seams. It redistributes them. Where consolidation is accompanied by a thinner regional posture, smaller staffs, or weaker liaison networks, it can create new vulnerabilities and new points of leverage for an adversary. In the Arctic, where distance, operational environment, and limited regional infrastructure can magnify the effects of delay or miscommunication, these vulnerabilities can become especially acute. These risks grow when reforms outpace the planning, exercising, and transition management needed to make them credible in practice. Sudden or sweeping change can also weaken cohesion at both the national and multinational levels—an outcome especially dangerous in an environment where adversaries seek to exploit not only geographic gaps, but also delays, ambiguities, and political hesitation within allied decision-making.

Resilience in such an environment cannot rest on connectivity alone. As Russia and China invest in capabilities designed to disrupt communications and target the nodes that enable joint and combined C2, resilience will depend on the ability to continue operating when networks are contested or communications are degraded. That requires redundancy: dispersed posture, hardened and backup communications, regional capacity, and command relationships that remain functional even when the system is under stress.

In that context, regional expertise allows commanders and planners to distinguish signal from noise, interpret ambiguous activity in a region where military, civilian, and commercial patterns often overlap, assess escalation pathways, and act with speed without compounding risk. As technology accelerates decision cycles, the need for speed should not supplant judgment: the ability to challenge assumptions, integrate perspectives across domains and theaters, and preserve options.

The same is true of contingency planning. In an environment where communications may be disrupted and warning time compressed, pre-planning must do more than catalogue scenarios.

It must connect national and allied authorities, bridge regional and functional divides, and account for how decisions will be made in contingencies—all while transcending classification levels, institutional boundaries, and public-private dependencies.

The goal is not to eliminate seams, which would be impossible, but to ensure that seams do not become strategic liabilities. For Arctic-relevant contingencies, the most credible end state is therefore not a seamless command architecture, but one designed to absorb friction, maintain political and military coherence, and deny adversaries clear opportunities to exploit physical, procedural, or cognitive gaps.

The views expressed herein are those solely of the authors. GMF as an institution does not take positions.

[1] There are varied definitions of the Arctic. The theater is most frequently described as the land and sea area above the Arctic Circle. US Code extends the definition with "all United States and foreign territory north of the Arctic Circle and all United States territory north and west of the boundary formed by the Porcupine, Yukon, and Kuskokwim Rivers [in Alaska]; all contiguous seas, including the Arctic Ocean and the Beaufort, Bering, and Chukchi Seas; and the Aleutian chain." For more definitions see Laura B. Comay, John Frittelli, Emma Kaboli, and Caitlin Keating-Bitonti, et al., "Changes in the Arctic: Background and Issues for Congress", July 14, 2025. https://www.congress.gov/crs-product/R41153

[2] "The High North" is a strategic term often used in defense policy debates with varying definitions. It is the English translation of the Norwegian nordområdene (meaning "the northern areas"), a political term coined in the 1970s. It is often understood as the European Arctic and its interlinked areas "roughly stretching from Greenland in the West to the Norwegian/Russia border in the Barents Sea in the East, and encompassing areas of strategic importance such as the Greenland-Iceland-UK (GIUK) Gap and Svalbard". UK Parliament, "On Thin Ice: UK Defence in the Arctic". https://publications.parliament.uk/pa/cm201719/cmselect/cmdfence/388/38805.htm

[3] Office of the Director of National Intelligence, "Annual Threat Assessment of the U.S Intelligence Community", March 14, 2026. https://www.dni.gov/files/ODNI/documents/assessments/ATA-2026-Unclassified-Report.pdf

[4] Department of War, "National Defense Strategy 2026", January 23, 2026. https://media.defense.gov/2026/Jan/23/2003864773/-1/-1/0/2026-NATIONAL-DEFENSE-STRATEGY.PDF

[5] Ibid.; "Congressional Commission on the U.S. Strategic Posture", Congressional Research Service, Updated February 12, 2025. https://www.congress.gov/crs_external_products/IF/PDF/IF12621/IF12621.4.pdf

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In the Arctic context, this produces a posture in which deterrence rests less on visible forward presence and more on integrated command structures, continuous situational awareness, and assured reinforcement pathways across domains.

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